Journal of Physical and Chemical Reference Data

Reference Data for the Density and Viscosity of Liquid Copper and Liquid Tin

Cite as: J. Phys. Chem. Ref. Data **39**, 033105 (2010); https://doi.org/10.1063/1.3467496
Submitted: 13 May 2010 . Accepted: 30 June 2010 . Published Online: 27 August 2010

Marc J. Assael, Agni E. Kalyva, Konstantinos D. Antoniadis, R. Michael Banish, Ivan Egry, Jiangtao Wu, Erhard Kaschnitz, and William A. Wakeham


View Online

Export Citation

ARTICLES YOU MAY BE INTERESTED IN

Reference Data for the Density and Viscosity of Liquid Aluminum and Liquid Iron
Journal of Physical and Chemical Reference Data **35**, 285 (2006); https://doi.org/10.1063/1.2149380

Reference Data for the Density and Viscosity of Liquid Cadmium, Cobalt, Gallium, Indium, Mercury, Silicon, Thallium, and Zinc
Journal of Physical and Chemical Reference Data **41**, 033101 (2012); https://doi.org/10.1063/1.4729873

A simple accurate method for measuring viscosity of liquid metals at high temperatures
Journal of Rheology **56**, 941 (2012); https://doi.org/10.1122/1.4709431



AIP Publishing

© 2010 American Institute of Physics.

Reference Data for the Density and Viscosity of Liquid Copper and Liquid Tin

Marc J. Assael,[a] **Agni E. Kalyva, and Konstantinos D. Antoniadis**
Chemical Engineering Department, Aristotle University, 54124 Thessaloniki, Greece

R. Michael Banish
Department of Chemical and Materials Engineering, University of Alabama in Huntsville, Huntsville, Alabama 35899, USA

Ivan Egry
Deutsches Zentrum für Luft- und Raumfahrt, Institut für Materialphysik im Weltraum, 51170 Köln, Germany

Jiangtao Wu
Center of Thermal and Fluid Science, School of Energy and Power Engineering, Xi'an Jiaotong University, Shaanxi 710049, People's Republic of China

Erhard Kaschnitz
Österreichisches Gießerei-Institut, Parkstraße 21, 8700 Leoben, Austria

William A. Wakeham
Department of Chemical Engineering, Imperial College, London SW7 2BY, United Kingdom

(Received 13 May 2010; accepted 30 June 2010; published online 27 August 2010)

The available experimental data for the density and viscosity of liquid copper and tin have been critically examined with the intention of establishing a density and a viscosity standard. All experimental data have been categorized into primary and secondary data according to the quality of measurement specified by a series of criteria. The proposed standard reference correlations for the density of copper and tin are characterized by standard deviations of 1.3% and 1.0% at the 95% confidence level, respectively. The standard reference correlations for the viscosity of copper and tin are characterized by standard deviations of 6.3% and 20% at the 95% confidence level, respectively. © *2010 American Institute of Physics*. [doi:10.1063/1.3467496]

Key words: copper; density; melt; reference data; tin; viscosity.

CONTENTS

1. Introduction. 2
2. Primary and Secondary Data. 2
3. Density. 2
   3.1. Experimental techniques. 2
   3.2. Data compilation. 3
   3.3. Density reference correlation. 4
4. Viscosity. 4
   4.1. Experimental techniques. 4
   4.2. Data compilation. 5
   4.3. Viscosity reference correlation. 5
5. Conclusions. 7
6. Acknowledgments. 7

List of Tables

1. Data sets considered for the density of liquid copper. 3
2. Data sets considered for the density of liquid tin. 3
3. Recommended values for the density of liquid copper and liquid tin. 5
4. Data sets considered for the viscosity of liquid copper. 6
5. Data sets considered for the viscosity of liquid tin. 6

[a]Author to whom correspondence should be addressed. Electronic mail: assael@auth.gr
© 2010 American Institute of Physics.

6. Recommended values for the viscosity of liquid copper and liquid tin. 8

List of Figures

1. Primary density data for liquid copper as a function of temperature. 4
2. Percentage deviations of the primary density data from Eq. (1) for liquid copper as a function of temperature. 4
3. Primary density data for liquid tin as a function of temperature. 4
4. Percentage deviations of the primary density data from Eq. (2) for liquid tin as a function of temperature. 5
5. Primary viscosity data for liquid copper as a function of temperature. 7
6. Percentage deviations of the primary viscosity data from Eq. (3) for liquid copper as a function of temperature. 7
7. Primary viscosity data for liquid tin as a function of temperature. 7
8. Percentage deviations of the primary viscosity data from Eq. (4) for liquid tin as a function of temperature. 7

1. Introduction

There is an increasing use of mathematical models to simulate a variety of processes involving liquid metals, such as shape-casting, primary and secondary metal productions, powder production by spray forming, and welding. Depending on which aspect of the process is modeled, there is a need for viscosity and density data for the relevant alloys. Historically, there are wide discrepancies in the viscosity data reported for the metallic elements and alloys.[1] For example, there is a spread of about 400% in the reported values of the viscosity for molten aluminum and about 100% for molten iron. For these reasons, a project was initiated by the International Association for Transport Properties (IATP) [former Subcommittee on Transport Properties of the International Union of Pure and Applied Chemistry (IUPAC)] to critically evaluate the density and the viscosity of selected liquid metals. Thus, in 2006, recommended values for the density and the viscosity of liquid aluminum and iron were published[2] as a result of a project supported by IUPAC. The current paper is a continuation of that work, and recommended values for the density and viscosity for liquid copper and tin are proposed. This work has also been carried out under the auspices of IATP and is supported by IUPAC.

2. Primary and Secondary Data

According to the recommendation adopted by the Subcommittee of Transport Properties (now known as The International Association for Transport Properties) of the International Union of Pure and Applied Chemistry, experimental data can be placed into two categories according to the quality of the data: primary and secondary data. The primary data are identified by the following criteria.[3]

(i) Measurements must have been made with a primary experimental apparatus, i.e., one for which a complete working equation is available.
(ii) The form of the working equation should be such that sensitivity of the property measured to the principal variables does not magnify the random errors of measurement.
(iii) All principal variables should be measurable to a high degree of precision.
(iv) The published work should include some description of purification methods and a guarantee of the purity of the sample.
(v) The data reported must be unsmoothed data. While graphs and fitted equations are useful summaries for the reader, they are not sufficient for standardization purposes.
(vi) The lack of accepted values of the viscosity of standard reference materials implies that only absolute and not relative measurement results can be considered.
(vii) Explicit quantitative estimates of the uncertainty of reported values should be given, taking into account the precision of experimental measurements and possible systematic errors.
(viii) Owing to the desire to produce low-uncertainty reference values, limits must be imposed on the uncertainty of the primary data sets. These limits are determined after critical evaluation of the existing data sets.

These criteria have been successfully employed to propose standard reference values for the viscosity and thermal conductivity of fluids over a wide range of conditions, with uncertainties in the vicinity of 1%.

In the case, however, of the liquid metals, it was felt and agreed that these criteria needed to be relaxed slightly, especially since the uncertainty of the measurements is much higher, primarily due to (i) the difficulties associated with the techniques employed at such high temperatures and (ii) the purity of the liquid metal sample which can be strongly affected by the surrounding atmosphere and the container used for the melt.

3. Density

3.1. Experimental techniques

Within the experimental data uncovered for the density of molten materials, a large number of techniques have been employed to measure the density of molten copper and tin. Methods employed include Archimedean, levitation, bubble-pressure, gamma radiation attenuation, pycnometer, and the capillary. These methods have been presented in our previous compilation,[2] and will not be discussed here.

The most commonly employed technique for the measurement of the density is the Archimedean technique. In this method, a solid sinker of known weight in air is suspended

TABLE 1. Data sets considered for the density of liquid copper

Reference	Publ. year	Purity (mass %)	Technique employed[a]	Uncertainty quoted (%)	Form of data[b]
Primary Data					
Brillo and Egry[4]	2004	—	Levitation (Abs)	1.0	D
Nasch and Steinemann[5]	1995	99.98	γ-Rays (Abs)	0.75	E
Bian *et al.*[6]	1986	—	Archimedean (Abs)	0.4	D
Watanabe and Saito[7]	1971	—	Bubble Pressure (Abs)	0.7	E
Frohberg and Weber[8]	1964	99.97	Bubble Pressure (Abs)	—	D
Lucas[9]	1964	—	Bubble Pressure (Abs)	—	P
Cahill and Kirshenbaum[10]	1962	99.972	Archimedean (Abs)	0.05	D
Lucas[11]	1961	—	Archimedean (Abs)	—	D
Secondary Data					
El-Mehairy and Ward[12]	1963	—	Levitation (Rel)	—	D

[a]Abs=absolute; Rel=relative.
[b]D=diagram; E=equation; P=points.

by a wire attached to the arm of a balance. When the sinker is entirely immersed in the liquid metal specimen, an apparent loss of weight is observed, arising mainly from the buoyant force exerted by the liquid metal sample. The loss of weight is simply related to the density of the liquid of immersion. Another very common method is the maximum-bubble-pressure technique, which is based on the formation of a hemispherical bubble of an inert gas at the tip of a capillary tube immersed to a certain depth in a liquid.

3.2. Data compilation

Tables 1 and 2 present the data sets found for the measurement of the density of liquid copper and liquid tin, respectively. In these tables, the purity of the sample, the technique employed, and the uncertainty quoted are also presented. Furthermore, the form in which the data are presented is noted. The data sets have been classified into primary and secondary sets according to the criteria in Sec. 2 and in conjunction with the techniques described in Sec. 3.1. More specifically, following the brief presentation of the various techniques employed for the measurement of the density of the two liquid metals, the following can be noted.

The Archimedean technique was successfully employed by Cahill and Kirshenbaum,[10] Lucas,[11] Wang *et al.*,[14] Bian *et al.*,[6] Kanda and Falkiewicz,[16] Berthou and Tougas,[18] and Kanda and Colburn.[19] In these papers, the technique is well described and these data sets are considered as primary data sets.

A pycnometer was employed very successfully by Thresh[20] and this set was included in the primary data set.

The maximum-bubble-pressure technique was successfully employed by Watanabe and Saito,[7] Frohberg and Weber,[8] Yatsenko *et al.*,[17] Lucas,[9] and Fisher and Phillips.[22] These data sets are thus considered as primary data sets. The

TABLE 2. Data sets considered for the density of liquid tin

Reference	Publ. year	Purity (mass %)	Technique employed[a]	Uncertainty quoted (%)	Form of data[b]
Primary Data					
Stankus and Kaiirulin[13]	2006	99.992	γ-Rays (Abs)	0.3	P
Wang *et al.*[14]	2003	99.9	Archimedean (Abs)	0.05	D
Alchagirov and Chochaeva[15]	2000	—	Twin Capillaries (Abs)	—	P
Nasch and Steinemann[5]	1995	99.98	γ-Rays (Abs.)	0.75	E
Bian *et al.*[6]	1986	—	Archimedean (Abs)	0.4	D
Kanda and Falkiewicz[16]	1973	99.999	Archimedean (Abs)	—	P
Yatsenko *et al.*[17]	1972	—	Bubble Pressure (Abs)	—	D
Berthou and Tougas[18]	1968	99.999	Archimedean (Abs)	0.02	E
Kanda and Colburn[19]	1968	—	Archimedean (Abs)	0.5	P
Thresh[20]	1968	99.99	Pycnometer (Abs)	0.1	D
Lucas[9]	1964	—	Bubble Pressure (Abs)	—	D
Herczynska[21]	1960	—	Capillary (Abs)	0.05	D
Fisher and Phillips[22]	1954	99.98	Bubble Pressure (Abs)	—	P
Secondary Data					
Friedrichs *et al.*[23]	1997	—	Bubble Pressure (-)	—	D

[a]Abs=absolute; Rel=relative.
[b]D=diagram; E=equation; P=points.

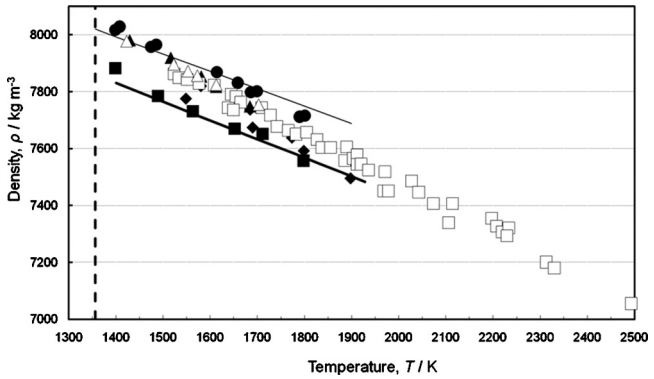


FIG. 1. Primary density data for liquid copper as a function of temperature: ■, Brillo and Egry (Ref. 4); –, Nasch and Steinemann (Ref. 5); ●, Bian *et al.* (Ref. 6); –, Watanabe and Saito (Ref. 7); ♦, Frohberg and Weber (Ref. 8); ▲, Lucas (Ref. 9); □, Cahill and Kirshenbaum (Ref. 10); △, Lucas (Ref. 11); - -, melting point.

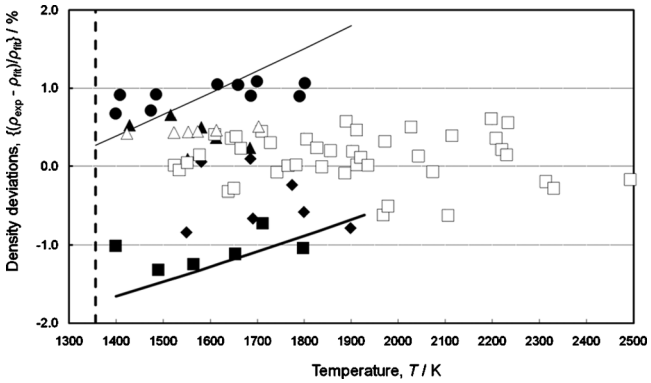


FIG. 2. Percentage deviations of the primary density data from Eq. (1) for liquid copper as a function of temperature: ■, Brillo and Egry (Ref. 4); –, Nasch and Steinemann (Ref. 5); ●, Bian *et al.* (Ref. 6); –, Watanabe and Saito (Ref. 7); ♦, Frohberg and Weber (Ref. 8); ▲, Lucas (Ref. 9); □, Cahill and Kirshenbaum (Ref. 10); △, Lucas (Ref. 11); - -, melting point.

measurements of Friedrichs *et al.*[23] showed a very large spread and they deviated too far from the rest of the data. Hence this data set was not considered as a primary data set.

The levitation technique was successfully employed by Brillo and Egry[4] in 2004 and this set was considered as a primary data set. The same method was employed by El-Mehairy and Ward[12] but in a relative manner, and they did not describe fully the technique employed. Hence this data set was not considered as a primary data set.

The gamma radiation attenuation technique was employed by Nasch and Steinemann[5] and Stankus and Kaiirulin.[13] Although there were some initial reservations in relation to the mass attenuation coefficient, both sets were finally considered as primary data sets.

Finally, the capillary technique was employed by Alchagirov and Chochaeva[15] and by Herczynska,[21] and both of these sets of data were considered as primary data sets.

3.3. Density reference correlation

The primary density data for liquid copper and tin, shown in Tables 1 and 2, respectively, were employed in a linear regression analysis to represent the density at 0.1 MPa as a function of the temperature. Since the quoted uncertainties were of similar magnitude, the data were weighted only according to the number of points. The following equations were obtained for the density ρ(kg m^{-3}) as a function of the absolute temperature T(K).

For copper (temperature range 1356–2500 K),

$$\rho = c_1 - c_2(T - T_{\text{ref}}), \tag{1}$$

where $c_1 = 7997$ kg m^{-3}, $c_2 = 0.819$ kg m^{-3} K^{-1}, and $T_{\text{ref}} = 1357.77$ K is the melting point of copper.[24] The standard deviation of Eq. (1) at the 95% confidence level is 1.3%.

For tin (temperature range 506–1950 K),

$$\rho = c_3 - c_4(T - T_{\text{ref}}), \tag{2}$$

where $c_3 = 6979$ kg m^{-3}, $c_4 = 0.652$ kg m^{-3} K^{-1}, and $T_{\text{ref}} = 505.08$ K is the melting point of tin.[24] The standard deviation of Eq. (2) at the 95% confidence level is 1.0%.

In Figs. 1–4, the primary data and their percentage deviations from Eqs. (1) and (2) are shown for copper and tin, respectively. The deviations are consistent with the quoted uncertainties. Density values calculated from Eqs. (1) and (2) are listed in Table 3.

4. Viscosity

4.1. Experimental techniques

There are many methods to measure the viscosity of liquids, but those suitable for liquid metals are limited by the low viscosities of metals (of the order of 1–10 mPa s), their chemical reactivity, and generally high melting points. Proposed methods include capillary, oscillating cup, rotational bob, oscillating plate, draining vessel, levitated drop, and acoustic methods. These methods have been presented in our previous compilation,[2] and will not be discussed here.

Most measurements use some form of oscillating cup viscometer. A vessel, normally a cylinder, containing the test

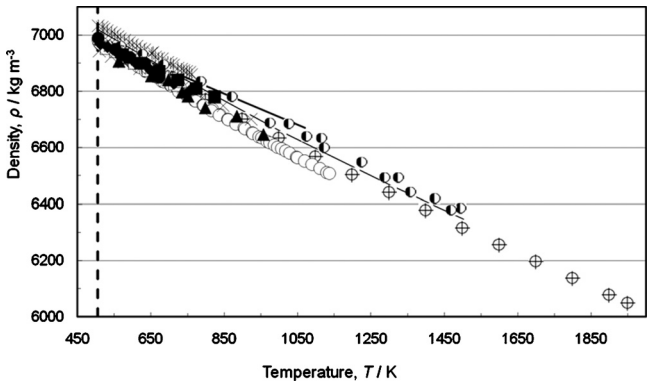


FIG. 3. Primary density data for liquid tin as a function of temperature: ⊕, Stankus and Kaiirulin (Ref. 13); ○, Wang *et al.* (Ref. 14); ⋇, Alchagirov and Chochaeva (Ref. 15); –, Nasch and Steinemann (Ref. 5); ◐, Bian *et al.* (Ref. 6); □, Kanda and Falkiewicz (Ref. 16); ×, Yatsenko *et al.* (Ref. 17); –, Berthou and Tougas (Ref. 18); △, Kanda and Colburn (Ref. 19); ♦, Thresh (Ref. 20); ▲, Lucas (Ref. 9); ■, Herczynska (Ref. 21); ●, Fisher and Phillips (Ref. 22); - -, melting point.

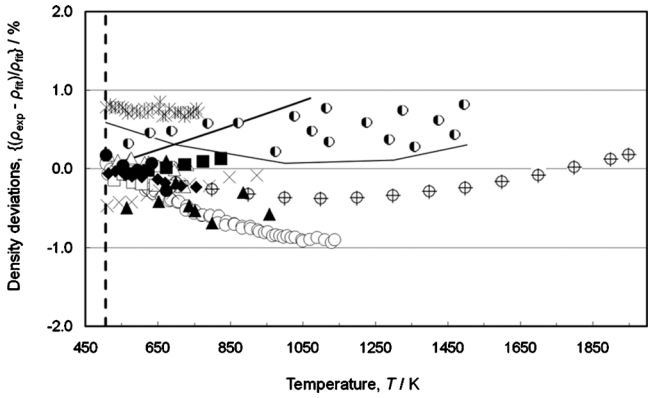


FIG. 4. Percentage deviations of the primary density data from Eq. (2) for liquid tin as a function of temperature: ⊕, Stankus and Kaiirulin (Ref. 13); ○, Wang *et al.* (Ref. 14); ⊁, Alchagirov and Chochaeva (Ref. 15); –, Nasch and Steinemann (Ref. 5); ◐, Bian *et al.* (Ref. 6); □, Kanda and Falkiewicz (Ref. 16); ×, Yatsenko *et al.* (Ref. 17); –, Berthou and Tougas (Ref. 18); △, Kanda and Colburn (Ref. 19); ♦, Thresh (Ref. 20); ▲, Lucas (Ref. 9); ■, Herczynska (Ref. 21); ●, Fisher and Phillips (Ref. 22); - -, melting point.

liquid is suspended by a torsion wire and is set in motion about the vertical axis. The oscillatory motion is damped by viscous friction within the liquid, and consequently, the viscosity is determined from the decrement and time period of the motion. This method was applied by Kehr *et al.*[25] in 2007, up to temperatures of approximately 2000 K and with a sufficiently high sensitivity for viscosities down to 1 mPa s. Unfortunately, the working equation is implicit and must be solved numerically. It should be emphasized that, according to our previous work,[2] data sets employing the 1952 equation of Knappwost[26] have not been considered as primary data sets; only those employing equations based on the 1958 work of Roscoe[27] have been considered.

For copper, only the oscillating cup method has been applied to measurement, with one exception: Iida *et al.*[28] used a capillary rheometer. In the case of tin, with its lower melting point, the gas bubble technique[23] and the rotating bob[29] have also been employed.

4.2. Data compilation

Tables 4 and 5 present the data sets found for the measurement of the viscosity of liquid copper and liquid tin, respectively. As in the case of the density measurements, papers prior to 1930 were not considered because sample purity was disputed before that time. In these tables, for every data set, the sample purity, the technique employed, the uncertainty quoted, the form of the data presented, the material in contact with the sample, the density source, and the equation employed are also shown. The data sets have been classified into primary and secondary sets according to the criteria in Sec. 2 and in conjunction with the techniques described in Sec. 4.1.

In the case of the viscosity data sets and in relation to the discussion of Sec. 4.1, the following points can be noted.

For copper, the data of Mao *et al.*[36] and Geng *et al.*[37] were discarded as primary data because they deviated too far from the rest of the data and from each other, although they were measured at the same laboratory. They were, however, kept as secondary data.

By the same token, their data for tin have also been considered secondary, together with those of Yang *et al.*[44] and Yao *et al.*[47] In addition, the data of Culpin[46] were only classified as secondary because the author applied an inadequate working equation. The data of Nakajima[45] and the data of Fisher and Phillips[22] were also both considered as secondary data because the former employed the equation developed by Knappwost[26] and the latter an earlier version. Finally, the data of Lewis,[48] taken in 1936, did not qualify as primary due to their age, but were retained as secondary data.

TABLE 3. Recommended values for the density of liquid copper and liquid tin

Liquid copper		Liquid tin	
Temperature (K)	Density (kg m^{-3})	Temperature (K)	Density (kg m^{-3})
1356	7998	506	6978
1400	7962	550	6950
1450	7921	600	6917
1500	7881	650	6885
1550	7840	700	6852
1600	7799	750	6819
1650	7758	800	6787
1700	7717	850	6754
1750	7676	900	6722
1800	7635	950	6689
1850	7594	1000	6656
1900	7553	1050	6624
1950	7512	1100	6591
2000	7471	1150	6559
2050	7430	1200	6526
2100	7389	1250	6493
2150	7348	1300	6461
2200	7307	1350	6428
2250	7266	1400	6396
2300	7225	1450	6363
2350	7184	1500	6330
2400	7143	1550	6298
2450	7102	1600	6265
		1650	6233
		1700	6200
		1750	6167
		1800	6135
		1850	6102
		1900	6070
		1950	6037

4.3. Viscosity reference correlation

The primary viscosity data for liquid copper and tin, shown in Tables 4 and 5, respectively, were employed in a regression analysis as a function of the temperature. Especially in the viscosity of tin, between the different sets, there are substantial differences that are much larger than the

TABLE 4. Data sets considered for the viscosity of liquid copper

Reference	Publ. year	Purity (mass %)	Technique employed[a]	Uncertainty quoted (%)	Form of data[b]
Primary Data					
Kehr *et al.*[25]	2007	—	Oscillating Cup (Abs)	—	D
Brillo *et al.*[30]	2007	—	Oscillating Cup (Abs)	7.0	D
Iida *et al.*[28]	1975	99.9999	Capillary (Abs)	0.5	P
Schenck *et al.*[31]	1963	—	Oscillating Cup (Abs)	—	P
Cavalier[32]	1959	99.999	Oscillating Cup (Abs)	—	D
Barfield and Kitchener[33]	1955	—	Oscillating Cup (Abs)	2	D
Gebhardt and Kostlin[34]	1952	—	Oscillating Cup (Abs)	—	P
Gebhardt and Worwag[35]	1951	—	Oscillating Cup (Abs)	—	P
Secondary Data					
Mao *et al.*[36]	2007	99.999	Oscillating Cup (Abs)	5	D
Geng *et al.*[37]	2003	—	Oscillating Cup (Rel)	5	D

[a]Abs=absolute; Rel=relative.
[b]D=diagram; E=equation; P=points.

quoted uncertainties, and generally systematic. Since almost all the measurements are absolute, the implication is that the measurements have a higher precision than accuracy. Moreover, there is no reason to believe that any of the measurements are systematically more reliable than others. Hence, the data were weighted only according to the number of points. The following equations were obtained for the viscosity η(mPa s) as a function of the absolute temperature T(K).

For copper, the fit equation is (in the temperature range 1356–1970 K)

$$\log_{10}(\eta/\eta^{\circ}) = -a_1 + \frac{a_2}{T}, \tag{3}$$

where $\eta^{\circ}=1$ mPa s, $a_1=0.4220$, and $a_2=1393.4$ K. The standard deviation of Eq. (3) at the 95% confidence level is 6.3%.

For tin, the fit equation is (temperature range 506–1280 K)

TABLE 5. Data sets considered for the viscosity of liquid tin

Reference	Publ. year	Purity (mass %)	Technique employed[a]	Uncertainty quoted (%)	Form of data[b]
Primary Data					
Kehr *et al.*[25]	2007	—	Oscillating Cup (Abs)	—	D
Friedrichs *et al.*[23]	1997	—	Gas Bubble (Abs)	—	D
Walsdorfer *et al.*[38]	1988	—	Capillary (Abs)	—	P
Iida *et al.*[28]	1975	99.999	Capillary (Abs)	—	P
Kanda and Falkiewicz[16]	1973	—	Oscillating Cup (Abs)	—	P
Thresh and Crawley[39]	1970	—	Oscillating Cup (Abs)	0.5	P
Kanda and Colburn[19]	1968	—	Oscillating Cup (Abs)	1.0	P
Menz and Sauerwald[40]	1966	99.9	Double capillary (Abs)	—	D
Schenck *et al.*[31]	1963	—	Oscillating Cup (Abs)		P
Bastien *et al.*[41]	1963	—	Capillary (Abs)	2.0	D
Rothwell[42]	1961	99.99	Oscillating Sphere (Abs)	3.0	P
Cavalier[32]	1959	99.999	Oscillating Cup (Abs)	—	D
Gebhardt and Kostlin[43]	1957	99.99	Oscillating Cup (Abs)	—	P
Jones and Davies[29]	1957	99.997	Rotating Bob (Abs)	—	P
Secondary Data					
Mao *et al.*[36]	2007	99.98	Oscillating Cup (Abs)	5.0	D
Yang *et al.*[44]	2004	99.99	Oscillating Cup (Abs)	5.0	D
Geng *et al.*[37]	2003	—	Oscillating Cup (Rel)	5.0	D
Nakajima[45]	1976	—	Oscillating Cup	—	E
Culpin[46]	1957	—	Oscillating Sphere	—	D
Fisher and Phillips[22]	1954	—	Oscillating Cup	—	P
Yao[47]	1952	99.989	Oscillating Cup (Rel)	—	P
Lewis[48]	1936	99.5	Capillary (Rel)	—	P

[a]Abs=absolute; Rel=relative.
[b]D=diagram; E=equation; P=points.

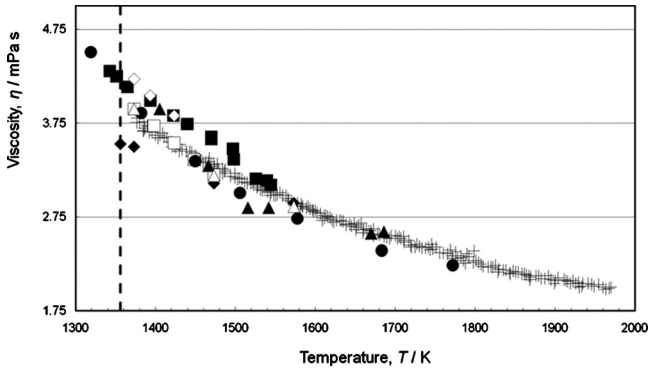


FIG. 5. Primary viscosity data for liquid copper as a function of temperature: +, Kehr *et al.* (Ref. 25); ■, Brillo *et al.* (Ref. 30); ◇, Iida *et al.* (Ref. 28); □, Schenck *et al.* (Ref. 31); ●, Cavalier (Ref. 32); ▲, Barfield and Kitchener (Ref. 33); ◆, Gebhardt and Kostlin (Ref. 34); △, Gebhardt and Worwag (Ref. 35); - -, melting point.

$$\log_{10}(\eta/\eta^{\circ}) = -a_3 + \frac{a_4}{T}, \tag{4}$$

where $\eta^{\circ}=1$ mPa s, $a_3=0.408$, and $a_4=343.4$ K. The standard deviation of Eq. (4) at the 95% confidence level is 20%.

In Figs. 5–8, the primary data and their percentage deviations from Eqs. (3) and (4) are shown for copper and tin, respectively. Viscosity values calculated from Eqs. (3) and (4) are given in Table 6.

5. Conclusions

The available experimental data for the density and viscosity of liquid copper and tin have been critically examined with the intention of establishing a density and a viscosity standard. All experimental data have been categorized into primary and secondary data according to the quality of measurement specified by a series of criteria. The proposed standard reference correlations for the density of the copper and tin are characterized by standard deviations of 1.3% and 1.0% at the 95% confidence level, respectively. The standard reference correlations for the viscosity of copper and tin are characterized by standard deviations of 6.3% and 20% at the 95% confidence level, respectively.

The proposed correlations are for saturation conditions. Although in some applications, such as the flow in a tube or a nozzle, the pressure is higher than the saturation pressure, the pressure dependences of the density and the viscosity of liquid metals are not sufficiently high for the variation to exceed the uncertainty in the correlations reported here.

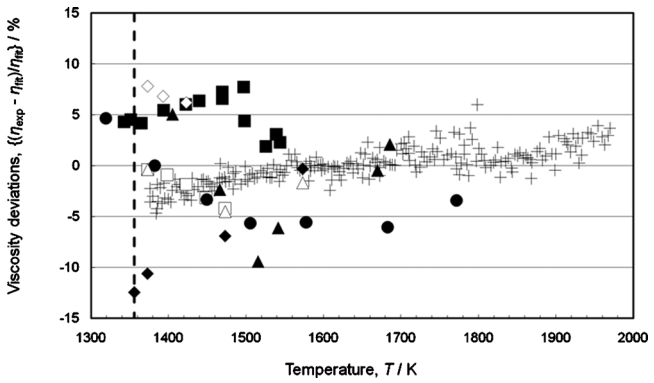


FIG. 6. Percentage deviations of the primary viscosity data from Eq. (3) for liquid copper as a function of temperature: +, Kehr *et al.* (Ref. 25); ■, Brillo *et al.* (Ref. 30); ◇, Iida *et al.* (Ref. 28); □, Schenck *et al.* (Ref. 31); ●, Cavalier (Ref. 32); ▲, Barfield and Kitchener (Ref. 33); ◆, Gebhardt and Kostlin (Ref. 34); △, Gebhardt and Worwag (Ref. 35); - -, melting point.

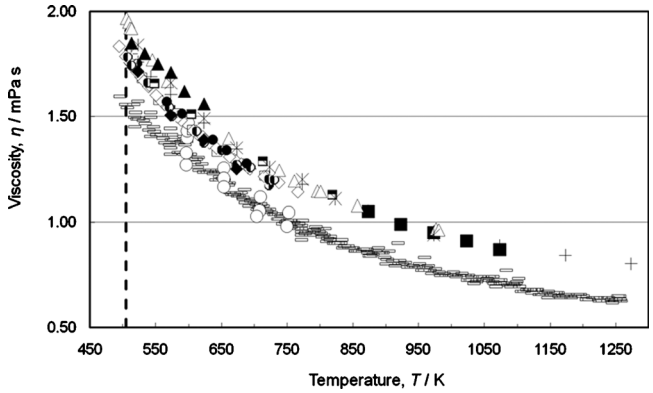


FIG. 7. Primary viscosity data for liquid tin as a function of temperature: –, Kehr *et al.* (Ref. 25); ○, Friedrichs *et al.* (Ref. 23); ◑, Walsdorfer *et al.* (Ref. 38); +, Iida *et al.* (Ref. 28); □, Kanda and Falkiewicz (Ref. 16); ◇, Thresh and Crawley (Ref. 39); ◐, Kanda and Colburn (Ref. 19); ●, Menz and Sauerwald (Ref. 40); ◆, Bastien *et al.* (Ref. 41); ■, Schenck *et al.* (Ref. 31); △, Rothwell (Ref. 42); ⬒, Cavalier (Ref. 32); ⚹, Gebhardt and Kostlin (Ref. 43); ▲, Jones and Davies (Ref. 29); - -, melting point.

6. Acknowledgments

The work described in this paper was carried out under the auspices of the International Association for Transport Properties (formerly known as the Subcommittee of Transport Properties of the International Union of Pure and Applied

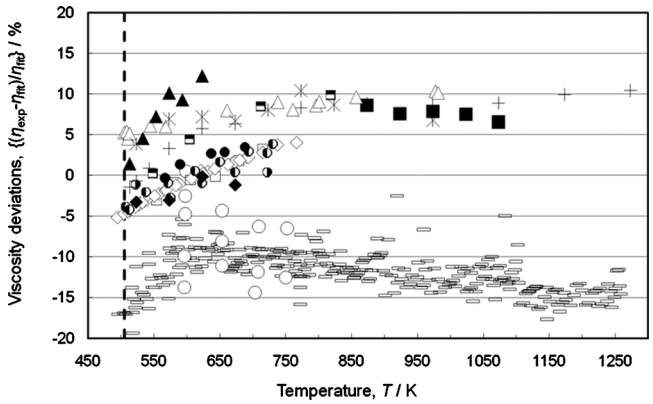


FIG. 8. Percentage deviations of the primary viscosity data from Eq. (4) for liquid tin as a function of temperature: –, Kehr *et al.* (Ref. 25); ○, Friedrichs *et al.* (Ref. 23); ◑, Walsdorfer *et al.* (Ref. 38); +, Iida *et al.* (Ref. 28); □, Kanda and Falkiewicz (Ref. 16); ◇, Thresh and Crawley (Ref. 39); ◐, Kanda and Colburn (Ref. 19); ●, Menz and Sauerwald (Ref. 40); ◆, Bastien *et al.* (Ref. 41); ■, Schenck *et al.* (Ref. 31); △, Rothwell (Ref. 42); ⬒, Cavalier (Ref. 32); ⚹, Gebhardt and Kostlin (Ref. 43); ▲, Jones and Davies (Ref. 29); - -, melting point.

TABLE 6. Recommended values for the viscosity of liquid copper and liquid tin

Liquid Copper		Liquid Tin	
Temperature (K)	Viscosity (mPa s)	Temperature (K)	Viscosity (mPa s)
1356	4.03	506	1.86
1400	3.74	550	1.65
1450	3.46	600	1.46
1500	3.21	650	1.32
1550	3.00	700	1.21
1600	2.81	750	1.12
1650	2.65	800	1.05
1700	2.50	850	0.99
1750	2.37	900	0.94
1800	2.25	950	0.90
1850	2.14	1000	0.86
1900	2.05	1050	0.83
1950	1.96	1100	0.80
		1150	0.78
		1200	0.76
		1250	0.74
		1280	0.72

Chemistry). The authors gratefully acknowledge the partial financial support of the International Union of Pure and Applied Chemistry.

[1] T. Iida and R. I. L. Guthrie, *The Physical Properties of Liquid Metals* (Clarendon, Oxford, 1988).
[2] M. J. Assael, K. E. Kakosimos, R. M. Bannish, J. Brillo, I. Egry, R. Brooks, P. N. Quested, K. C. Mills, A. Nagashima, Y. Sato, and W. A. Wakeham, J. Phys. Chem. Ref. Data **35**, 285 (2006).
[3] M. J. Assael, M. L. V. Ramires, C. A. Nieto de Castro, and W. A. Wakeham, J. Phys. Chem. Ref. Data **19**, 113 (1990).
[4] J. Brillo and I. Egry, Z. Metallkd. **95**, 691 (2004).
[5] P. M. Nasch and S. G. Steinemann, Phys. Chem. Liq. **29**, 43 (1995).
[6] M. S. Bian, L. M. Ma, and J. T. Wang, Acta Metall. Sin. **22**, B90 (1986).
[7] S. Watanabe and T. Saito, Nippon Kinzoku Gakkaishi **35**, 554 (1971).
[8] M. G. Frohberg and R. Weber, Arch. Eisenhuettenwes. **35**, 877 (1964).
[9] L. D. Lucas, Mem. Sci. Rev. Metall. **61**, 1 (1964).
[10] J. A. Cahill and A. D. Kirshenbaum, J. Inorg. Nucl. Chem. **66**, 1080 (1962).
[11] L. D. Lucas, Compt. Rend. **253**, 2526 (1961).
[12] A. E. El-Mehairy and R. G. Ward, Trans. Metall. Soc. AIME **227**, 1226 (1963).
[13] S. V. Stankus and R. A. Kaiirulin, High Temp. **44**, 389 (2006).
[14] L. Wang, Q. Wang, A. Xian, and K. Lu, J. Phys.: Condens. Matter **15**, 777 (2003).
[15] B. B. Alchagirov and A. M. Chochaeva, High Temp. **38**, 44 (2000).
[16] F. A. Kanda and M. J. Falkiewicz, High. Temp. Sci. **5**, 252 (1973).
[17] S. P. Yatsenko, V. I. Kononenko, and A. L. Suhman, High Temp. **10**, 55 (1972).
[18] P. E. Berthou and R. Tougas, J. Less-Common Met. **16**, 465 (1968).
[19] F. A. Kanda and R. P. Colburn, Phys. Chem. Liq. **1**, 159 (1968).
[20] H. R. Thresh, J. Inst. Met. **96**, 308 (1968).
[21] E. Herczynska, Naturwiss. **47**, 200 (1960).
[22] H. J. Fisher and A. Phillips, J. Met. **6**, 1060 (1954).
[23] H. A. Friedrichs, L. W. Ronkow, and Y. Zhou, Steel Res. **68**, 209 (1997).
[24] H. Preston-Thomas, Metrologia **27**, 3 (1990).
[25] M. Kehr, W. Hoyer, and I. Egry, Int. J. Thermophys. **28**, 1017 (2007).
[26] A. Knappwost, Z. Phys. Chem. **200**, 81 (1952).
[27] R. Roscoe, Proc. Phys. Soc. **72**, 576 (1958).
[28] T. Iida, Z.-I. Morita, and S. Takeuchi, J. Jpn. Inst. Met. **39**, 1169 (1975).
[29] W. R. D. Jones and J. B. Davies, J. Inst. Met. **86**, 164 (1957).
[30] J. Brillo, R. Brooks, I. Egry, and P. N. Quested, Int. J. Mater. Res. **98**, 457 (2007).
[31] H. Schenck, M. G. Frohberg, and K. Hoffmann, Arch. Eisenhuettenwes. **34**, 93 (1963).
[32] M. G. Cavalier, in *The physical chemistry of metallic solutions and intermetallic compounds*, Proceedings of a Symposium held at the National Physical Laboratory on the 4th, 5th, and 6th of June 1958 (H. M. Stationery Off., London, 1959), Vol. 2, paper 4D.
[33] R. N. Barfield and J. A. Kitchener, J. Iron Steel Inst., London **180**, 324 (1955).
[34] E. Gebhardt and K. Kostlin, Z. Metallkd. **43**, 292 (1952).
[35] E. Gebhardt and G. Worwag, Z. Metallkd. **42**, 358 (1951).
[36] T. Mao, X. F. Bian, X. Y. Xue, Y. N. Zhang, J. Guo, and B. A. Sun, Physica B **387**, 1 (2007).
[37] H.-X. Geng, H.-R. Geng, X.-Y. Xue, K. Yu, and J.-T. Liu, Chin. Phys. Lett. **20**, 1102 (2003).
[38] H. Walsdorfer, I. Arpshofen, and B. Predel, Z. Metallkd. **79**, 503 (1988).
[39] H. R. Thresh and A. F. Crawley, Metall. Trans. **1**, 1531 (1970).
[40] W. Menz and F. Sauerwald, Acta Metall. **14**, 1617 (1966).
[41] P. Bastien, P. Azou, and J.-C. Armbruster, Fonderie **204**, 43 (1963).
[42] E. Rothwell, J. Inst. Met. **90**, 389 (1961).
[43] E. Gebhardt and K. Kostlin, Z. Metallkd. **48**, 636 (1957).
[44] Z. X. Yang, H. R. Geng, Z. D. Tao, and C. J. Sun, J. Atom. Mol. Phys. **21**, 663 (2004).
[45] H. Nakajima, Trans. Jpn. Inst. Met. **17**, 403 (1976).
[46] M. F. Culpin, Proc. Phys. Soc. London, Sect. B **70**, 1069 (1957).
[47] T. P. Yao and V. Kondig, J. Inst. Met. **81**, 17 (1952).
[48] A. J. Lewis, Proc. Phys. Soc. London **48**, 102 (1936).